

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8495**

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 401	Seat No: 2022600050
Course Section: 1		

L_29_507_986_L_1505_38342



Name of the candidate. Aparna Sangepu S

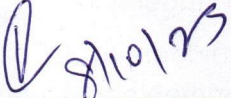
Candidate's UID number:

2	0	2	2	6	0	0	0	5	0
---	---	---	---	---	---	---	---	---	---

Examination class room number:

4	0	1
---	---	---

Invigilator's signature with date:



(To be filled in by the candidate)
EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)
PROGRAM: CSE-AIML
DAY AND DATE: Wednesday 08-10-25
COURSE NAME: Deep Learning

SEMESTER : 7
I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

Q.1b) i) The expected squared error of a predictor can be given as

$$ESE = \|y_{true} - y_{pred}\|^2$$

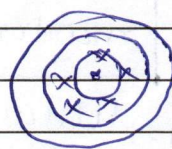
ii) When the bias of a predictor is high, it results in underfitting whereas a high variance covers a good amt. of training data and results in overfitting.

iii) The predictor has more training accuracy and less accuracy on testing data when overfitting occurs.

iv) This is the bias-variance decomposition.



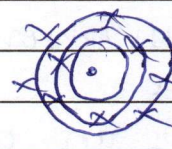
Bias ↓
variance ↓



Variance ↑
Bias ↓



Bias ↑
Variance ↓



Bias ↑
Variance ↑

v) Capacity of a model is defined as its ability to learn from the input data.

vi) A high capacity means a model can learn the complex relationships among the data whereas a low capacity model is only capable of capturing simple features.

vii) When a model capacity is too high, it has a very high variance which means it learns the patterns in the training data so well, it results in overfitting.

viii) Also since, the model is now well trained on the input data, it fails to capture the test data and the test accuracy is low, which affects generalization.

Space
For
MarksQuestion
No.

START WRITING HERE

(Begin answer for each question on a new page)

Q.2b) i) An autoencoder is a variant of artificial neural network (ANN) which reproduces its input after compressing it into a lower dimensional state.

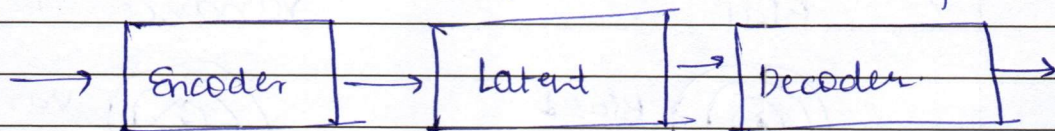
ii) It has three components:

a) Encoder

b) Latent space

c) Decoder

iii) The encoder compresses the input and extracts the features for further computation. The latent holds the reduced input and the decoder produces the output.



iv) A linear undercomplete ^{auto}encoder is an autoencoder where the latent space dimensions are less than the input size.

v) So an undercomplete autoencoder captures only the essential features and stores it in the latent space.

vi) This is a reduction of dimensions in an attempt to capture high variance features.

vii) A linear undercomplete autoencoder trained with MSE performs dimensionality reduction like in PCA.

viii) This creates a subspace equivalent to PCA where the principal components with high variance are captured.

ix) An overcomplete autoencoder has latent space dimensions greater than the input size to capture all the essential features.

x) This involves a drawback of copying the input as it is during compression.

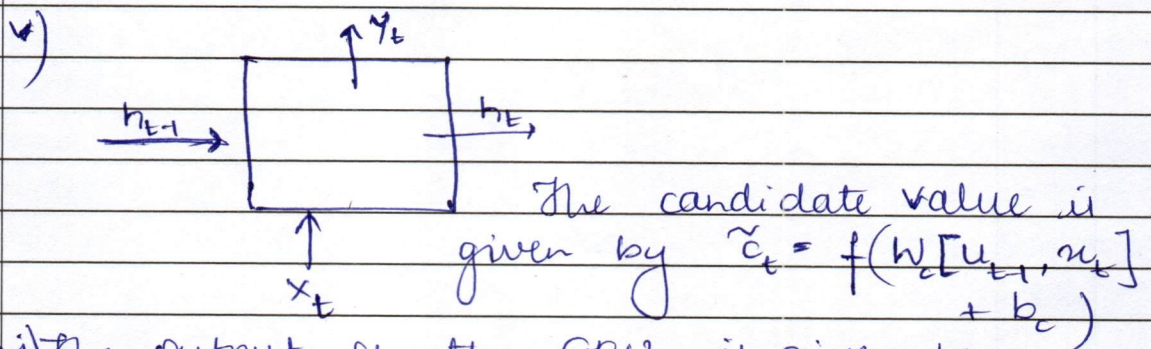
xi) Certain regularization need to be added to avoid these.

[illegible]

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

- Q3b) i) A Gated Recurrent Unit is a modified LSTM where a cell has one hidden state instead of two like in LSTM.
- ii) A GRU is capable of retaining both long and short term memory (just like an LSTM).
- iii) A GRU consists of two gates -
- Update gate
 - Reset gate
- iv) These gates control the information flow into the network. Their equations can be given as
- $$\begin{aligned} u_t &= \sigma(W_u[h_{t-1}, x_t] + b_u) \\ r_t &= \sigma(W_r[h_{t-1}, x_t] + b_r) \end{aligned}$$



vi) The output of the GRU is given by

$$z_t = W_c u_t + (1 - W_c) \tilde{c}_t + b_z$$

vii) Thus the update rule can be interpreted as a convex combination of hidden state and candidate state.

viii) LSTMs have two hidden states: the cell hidden state and the candidate hidden state. It consists of two gates - forget gate and input gate for the control of information flow.

ix) The equations are given as follows:

$$\begin{aligned} f_t &= \sigma(W_f(h_{t-1}, x_t) + b_f) \\ i_t &= \sigma(W_i(h_{t-1}, x_t) + b_i) \end{aligned}$$



Space
For
Marks

Question
No.

START WRITING HERE
(Begin answer for each question on a new page)

- x) LSTMs are thus tending to have more parameters in comparison to GRUs.
- xi) In the real-world scenario, GRUs can be efficiently used in place of LSTMs as they are less costly and equally computationally effective.

Sr. No.



Space
For
Marks

Question No.	Answer
1	1. The first step in the process of creating a new product is to identify a market need. This is often done through market research, which can involve surveys, focus groups, and other methods of gathering information about potential customers. Once a market need is identified, the next step is to develop a concept for a product that meets that need. This involves brainstorming ideas and selecting the most promising one. The third step is to create a prototype of the product, which allows the company to test the concept and make any necessary adjustments. Finally, the product is launched into the market, and the company monitors its performance and makes any necessary adjustments to the marketing strategy.

START WRITING HERE
(Begin answer for each question on a new page)

[illegible]

START WRITING HERE

(Begin answer for each question on a new page)

[illegible]

[illegible]

8495



(Begin answer for each question on a new page)

[illegible]



START WRITING HERE

START WRITING HERE
(Begin answer for each question on a new page)

[illegible]